

Efficient atmospheric water extraction unveiled

University of Utah engineering researchers have advanced atmospheric water harvesting (AWH) to aid billions facing water shortages. They developed a compact, fuel-fired



The prototype has achieved its target of producing 5 litres of water per day per kilogram of adsorbent material (Credit: <https://attheu.utah.edu>)

AWH device using aluminium fumarate, a metal-organic framework (MOF), to extract water from non-humid air. MOFs are ideal due to their high affinity for water and selective adsorption capabilities. The prototype uses aluminium fumarate panels to capture water molecules, boasting a vast internal surface area. Heat from an Army camping stove converts the absorbed

water into liquid. This efficient technology addresses military hydration needs in remote areas and offers a promising solution for civilian water scarcity, outperforming solar-powered systems in low humidity regions.

3D integrated e-skin advances health monitoring

Researchers at the Hong Kong Polytechnic University and other Chinese institutes developed a 3D integrated, flexible electronic skin, reported in Nature Electronics. This e-skin combines high-density inorganic components with organic stretchable substrates, surpassing traditional systems' limitations. It features multilayered liquid metal circuits and fibre mats, maintaining electrical properties up to 1500% strain and allowing air and moisture passage. The thin, durable material records and transmits physiological signals, enabling battery-powered, wireless bioelectronic devices for stable biosignal monitoring and electrostimulation.

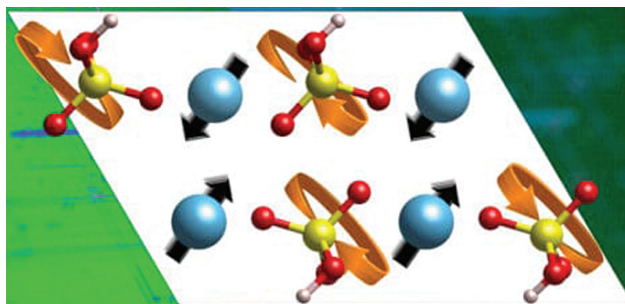
Lizard robot unveils optimal movement patterns

Researchers at the University of the Sunshine Coast have created a robot inspired by lizard locomotion to study the evolution and advantages of different movements in humans and animals. This lizard-like robot, capable of running and climbing, combines biology, robotics, and mathematics. The study found that optimal locomotion requires both spine

and limb movement, mirroring lizard patterns, and revealed evolutionary trade-offs between speed, efficiency, and stability. The robot's performance data, mapped against 40 lizard species, highlighted that lateral spine undulation reduces energy consumption, offering insights for designing efficient, versatile robotic systems for diverse environments.

Proton conductors unlock green energy potential

MIT engineers have advanced proton conductor development by identifying traits for rapid conduction. Through simulations and quantitative analysis, they discovered six promis-



A class of materials called solid acids were especially likely to be fast proton conductors, based on computer simulations of the materials' behaviour (Credit: Courtesy of researchers; MIT News)

ing materials that may outperform existing ones, pending experimental confirmation. Current hydrogen production methods emit significant CO₂. Efficient proton conductors in electrochemical systems could transform industrial chemical and fuel production, including ammonia. Most proton conductors operate at high temperatures (200-600°C), which are energy-intensive and degrading. The MIT team studied solid acids, known for conduction above 200°C, and identified key phonons for lattice flexibility. These findings could enhance electrochemical cells for producing hydrogen and ammonia, though practical applications are years away.

High-frequency RF switches accelerate communication

Researchers from King Abdullah University of Science and Technology, Universitat Autònoma de Barcelona, and other institutions have developed memristive RF switches operating at over 100GHz. These switches use memristors, which change resistance under stress, toggling between high and low resistance states. These switches use metal-insulator transition materials to address non-idealities and device parasitics for optimal performance. The team's multilayer h-BN switches enhance reliability and performance, supporting advanced communication technologies and faster networks like 5G and 6G.